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# The Effect of Leaded Gasoline on Elderly Mortality: Evidence from Regulatory Exemptions[†]

*By* Alex Hollingsworth and Ivan Rudik*

*Leaded gasoline is still used globally for aviation and automotive racing. Exploiting regulatory exemptions and a novel quasi-experiment, we find that leaded gasoline use in racing increases ambient lead, elevated blood lead rates, and elderly mortality. The mortality estimates indicate that each gram of lead added to gasoline exceeds $1,100 in damages. Our setting allows us to rule out potential confounders, such as correlated pollutants or socioeconomic status. We provide the first causal estimates linking adult mortality to leaded gasoline, highlight the value of banning on-road leaded gasoline, and present policy-relevant cost estimates at the lowest ambient levels to date.* (*JEL* I12, J14, L71, Q51, Q53)

Lead (Pb) is a toxic heavy metal associated with numerous adverse outcomes ranging from increased aggressive behavior to premature mortality. In response to mounting health concerns, countries around the world began taking action in the 1970s to reduce lead exposure. In 1975, Japan phased lead out of regular gasoline and the United States mandated all new model-year cars to have emission control devices that required unleaded instead of leaded gasoline (Newell and Rogers 2003, Yoshinaga 2012). Many European and Asian countries began lead phasedowns in the 1980s (Nriagu 1990), while African and Middle Eastern countries generally began the process in the 1990s or 2000s (Thomas 1995, Lovei 1998). As of 2020, leaded gasoline has been phased out of regular on-road use in all but one country (UN Environment Programme 2020). The reduction of lead in gasoline produced

*Hollingsworth: O'Neill School of Public and Environmental Affairs, Indiana University (email: hollinal@indiana.edu); Rudik: Charles H. Dyson School of Applied Economics and Management, Cornell University (email: irudik@cornell.edu). Lucas Davis was coeditor for this article. We thank Patrick Carlin, Livy Crim, Sam Kuster, and Shyam Raman for research assistance. We thank Lindsay Allen, Judd Boomhower, Karen Clay, Brian Dillon, Maoyong Fan, Teevrat Garg, Scott Grosse, Dave Keiser, Jonathan Ketcham, Cathy Kling, David Konisky, Gabe Lade, David Molitor, Julian Reif, Jessica Reyes, Nick Sanders, Edson Severnini, Kosali Simon, Gregory Steele, Coady Wing, Nikos Zirogiannis, Eric Zou, and seminar participants at the AERE Summer Conference, the Annual Health Economics Conference, the ASHEcon Annual Conference, Carnegie Mellon University, Cornell University, EPA NCEE, the Heartland Workshop, Indiana University, the Inter-American Development Bank, Miami University, the NBER EEE Spring Meetings, UC Davis, UC San Diego, the University of Connecticut, and the University of Louisville for valuable comments. A replication package for this project can be found at http://doi.org/10.3886/E120128V1 (Hollingsworth and Rudik 2021).

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steep declines in ambient lead concentrations, representing one of the greatest public health achievements in recent history.[1]

Despite these advances, the phaseout of lead from gasoline remains incomplete. While virtually every country has banned the sale of leaded gasoline for on-road use, it is still widely allowed for off-road uses such as aviation and automotive racing.[2] The continued use of lead remains a major public health concern as thousands of communities still grapple with lead contamination from legacy sources (Pell and Schneyer 2016), and research increasingly suggests there is no safe level of exposure (Centers for Disease Control and Prevention 2012, Lanphear 2017).

In this paper, we use the automotive racing exemption from the United States Clean Air Act to causally identify the external effects of leaded gasoline. We take advantage of a novel quasi-experiment where two major racing organizations—the National Association for Stock Car Auto Racing (NASCAR) and the Automobile Racing Club of America (ARCA)—switched from leaded to unleaded gasoline in 2007.[3] Using difference-in-differences and event study designs, we estimate the relationship between leaded gasoline use and each link in the causal chain from release of lead in the air, to accumulation of lead in blood, and finally to impacts on human mortality. We find that each leaded race significantly increases nearby ambient lead concentrations in the weeks following the race. We further find that leaded races increase the prevalence of elevated blood lead levels (EBLLs) in children and increase elderly mortality. This provides the first causal evidence that leaded gasoline exposure increases adult mortality.

We estimate that the reduction in annual lead emissions from deleading NASCAR and ARCA races yielded social benefits of $1,800 per child in counties with racetracks, and $600 per child in bordering counties through avoided IQ reductions. The benefits from avoided premature elderly mortality in racetrack and bordering counties are over $15 million per race, totaling $2.2 billion per year. These annual benefits are more than the estimated value of all NASCAR teams combined (Smith 2020).

The mortality effects alone put a lower bound on the social cost of a gram of lead in gasoline at $1,100. These magnitudes, while large, are consistent with prior observational findings on the relationship between adult mortality and lead exposure (Pirkle et al. 1985, Lustberg and Silbergeld 2002, Menke et al. 2006, Lanphear et al. 2018). For comparison, the equivalent mortality reduction from deleading NASCAR and ARCA races could be generated by reducing $PM_{2.5}$ concentrations in the areas surrounding racetracks by 11 $\mu g/m^3$, 90 percent of 2006 levels (Deryugina et al. 2019).

[1] Lead was used in a variety of products because of its efficacy and versatility. It was added to gasoline for its lubrication properties and to boost octane, reducing the likelihood of preignition; it was added to paint to increase moisture resistance, brightness, and durability, as well as to accelerate drying; and it has been used in pipes and pipe solder because of its malleability.

[2] In the United States, federal regulations allow unleaded gasoline to contain up to 0.05 grams of lead per gallon. This is about 2 percent of the lead content added to gasoline in the 1970s. See 40 CFR 80.140 and 40 CFR 80.173 (b).

[3] The EPA had been concerned about racing lead emissions leading up to 2007. A 2006 report on lead writes, "As was the case with on-road emissions during the time of universal leaded gasoline use, the combustion of racing fuel likely elevates ambient Pb concentrations in the nearby area. This may pose some health risk for some subpopulations, such as residents living in the vicinity of racetracks, fuel attendants, racing crew and staff, and spectators." (US Environmental Protection Agency 2006).

The deleading of automotive races is an ideal setting to identify the effects of lead since it overcomes identification challenges such as residential sorting and correlated declines in emission of lead and other pollutants at the tailpipe. The switch to unleaded fuel was the only fuel-relevant change made by NASCAR or ARCA in 2007, and races continued in the same places and at roughly the same times of year. In our empirical approach, we compare changes in population outcomes, before and after deleading, between those who live near racetracks and those who live further away. This enables within-population comparisons that alleviate problems with persistent confounders such as socioeconomic status, the stock of soil lead, the presence of leaded household paint, and lead emissions from nearby municipal airports.

Races emit large quantities of lead that significantly increase ambient concentrations, providing sufficient statistical power for our analysis. Figure 1 compares the magnitude of lead emissions from races to the distribution of annual airborne lead emissions at Toxic Release Inventory (TRI) facilities, and to annual lead emissions at airports. A single three-hour race emits more lead than the annual total of 70 percent of all lead-emitting facilities in the TRI, and as much lead as the average airport in an entire year. If all races were aggregated into a single TRI entity they would emit an estimated 2 million grams of lead per year, collectively ranking as the thirtieth largest emitter out of over 5,000 facilities and accounting for roughly half a percent of all annual US lead emissions.[4,5]

This work makes several contributions to the literature. The main contribution is that we use a quasi-experiment to causally link three economically relevant outcomes to well-defined sources and quantities of leaded gasoline. In turn, we provide the first causal evidence of the effects of leaded gasoline exposure on adult mortality and construct a credible lower bound on the social cost of lead in gasoline. Second, the deleading of racing fuel allows us to separately identify the effect of lead emissions from other correlated pollutants, socioeconomic factors, and legacy lead in the soil and on the road. This is a critical distinction between prior research designs and our approach. Third, we show that the effects of deleading are readily apparent in the raw data, and we use the same study design to estimate the effects on each link in the emission-outcome chain, assuring internal validity. Fourth, our work sheds light on the adverse effects of lead at some of the lowest levels of ambient lead exposure studied in the literature.[6] This allows us to provide policy-relevant estimates that are pertinent to modern exposure levels. A nascent literature, to which this paper contributes, increasingly demonstrates that there is no safe level of lead exposure (Aizer et al. 2018, Grönqvist et al. 2018, Lanphear et al. 2018).

This paper also helps to quantify both the value of the global deleading of on-road gasoline and the large social costs of continued usage of leaded gasoline for racing

[4] This is 800 times the median annual TRI facility emissions and 25 times the average.

[5] Prior research has demonstrated that employees on race teams (e.g., pit crews, mechanics) had high blood lead levels prior to the 2007 deleading. Blood lead levels for employees on NASCAR race teams in 1968 and 2006 were over 7 $\mu$g/dl higher than the population averages in those same years. Testing in 2006 showed that 40 percent of NASCAR team members had blood lead levels above 10 $\mu$g/dl, the threshold at the time for elevated blood lead (Polacek and Collins 1968, O'Neil et al. 2006).

[6] One caveat to our findings is that the same quantity of emissions may differentially affect health if released over a longer time period. Short-lived but high-rate exposure (e.g., an automotive race) may affect health differently than constant but low-rate exposure (e.g., industrial facility).

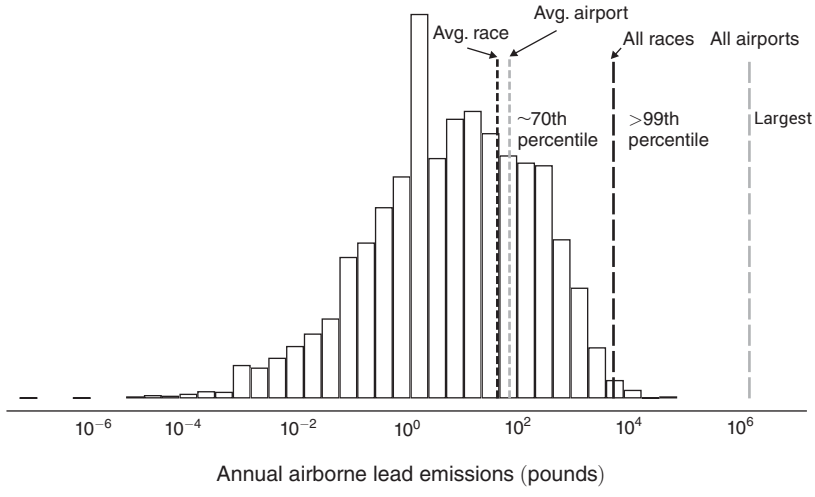



FIGURE 1. AUTOMOTIVE RACING AND AVIATION LEAD EMISSIONS COMPARED TO DISTRIBUTION OF ALL LEAD-EMITTING FACILITIES IN THE 2005 TRI

*Notes:* Toxic Release Inventory (TRI) facility emissions are the sum of fugitive and stack emissions in 2005, the year prior to unleaded race tests. The distribution is similar for 2006. We only include TRI facilities with positive quantities of airborne lead emissions. TRI and airport emissions are annual totals while average race emissions are about a three-hour total. Airborne lead emissions from both automotive races and aircraft are not reported in the TRI. We obtain the amount of lead emitted from NASCAR and ARCA races using an estimate of 3.2 race miles per gallon as the average fuel economy. The methodology behind the fuel economy estimate can be found in online Appendix Section A.10, and total emissions are calculated in online Appendix Section A.11.

and aviation. While NASCAR and ARCA now use unleaded fuel, leaded racing fuel can still be legally purchased at hundreds of locations across the United States and remains widely used in both professional and amateur races. Our mortality cost estimate indicates that the efficient price for leaded racing fuel is orders of magnitude higher than current prices, which range from $10–$20 per gallon. The continued use of leaded racing fuel puts participants, attendees, and those who live nearby at risk of bearing undue external costs. Although we do not directly study aviation gasoline, our results are suggestive that the use of leaded fuel by piston-engine aircraft remains a major threat to public health worldwide. Over a quarter of a million aircraft across the world still use leaded avgas (Federal Aviation Administration 2020), and over 500,000 gallons are used each day in the United States alone (Kessler 2013).

The paper proceeds as follows. Section I describes the physiology of lead and relevant prior research. Section II reports data sources. Section III contains methods and results. Section IV computes the value of external effects of leaded gasoline on blood lead levels and premature mortality. Section V concludes. The online Appendix contains the history of NASCAR and ARCA's switch to unleaded fuel, information on the ambient lead sample, information on other events at tracks, an additional cross-sectional soil analysis, information on collinearity issues, summary statistics, robustness checks and an alternative split-sample instrumental variables approach, associations related to environmental justice, a balance table, results from a survey testing knowledge about NASCAR fuel use, an out-of-sample validation of our ambient air estimates, a counterfactual prediction of ambient lead levels if

leaded gas was still used for on-road driving, and additional calculations used to put our effects into context.

## I. The Physiology of Lead and Prior Work

Lead can enter the body through the skin, digestive tract, or by inhalation.[7] Once in the blood stream, lead can be deposited in both soft (e.g., liver and brain) and hard (e.g., teeth and bone) tissues. Lead has a half life of approximately 30 days in the blood, up to a few years in soft tissues, and several decades in hard tissues. It serves no purpose in the human body, however it competes with and displaces necessary metals such as calcium, zinc, and magnesium. This displacement has a number of negative physiological consequences ranging from impaired cellular signaling to cellular death.

Tetraethyl lead (TEL) is the primary lead compound added to gasoline. TEL boosts octane, reducing the likelihood of pre-ignition of the gasoline. TEL is particularly toxic, even when compared to forms of lead historically found in household paint (Agency for Toxic Substances and Disease Registry 2018). TEL is easily absorbed through the skin and other major routes of entry, and it can also cross the blood-brain barrier (Abadin and Pohl 2010). These properties make it extremely harmful at low levels of exposure.[8]

A large body of prior work has studied the correlation between lead exposure and adverse health outcomes. While lead affects nearly every organ in the body, the most well-known impacts are on the nervous system. This is particularly salient for young children, where long run IQ declines are associated with early life exposure (Baghurst et al. 1992, Lanphear et al. 2005). Children are more sensitive to lead because, conditional on exposure, they absorb more lead than adults and because their nervous systems are still developing (Lidsky and Schneider 2003).

Lead is also associated with increased cardiovascular ailments including hypertension, and higher rates of plaque build up and clotting in blood vessels. Jointly these cardiovascular issues serve to increase the risk of ischemic heart disease and death (Lanphear et al. 2018). Previous research has found strikingly large estimates of the association between lead exposure and adult mortality. Lanphear et al. (2018) estimate that over 400,000 (18 percent) deaths per year in the United States are associated with lead exposure; Pirkle et al. (1985) find that the decrease in blood lead levels (BLLs) in the late 1970s was associated with a 6 percent decrease in all-cause mortality; Lustberg and Silbergeld (2002) find that having BLLs in the range of 10–19 $\mu$g/dl is associated with a 17 percent increase in the risk of all-cause mortality; and Menke et al. (2006) find that having BLLs above 3 $\mu$g/dl is associated with a 30 percent increase in all-cause mortality. These correlational studies have generally found the strongest mortality associations with cardiovascular-related causes of death.

[7] See Papanikolaou et al. (2005); Garza, Vega, and Soto (2006); and US Environmental Protection Agency (2006) for more details on how it enters and affects the body. See Ranft et al. (2008) for a review of how ambient lead concentrations correlate more strongly with blood lead levels than soil lead concentrations. See He, Wang, and Zhang (2009) and Oh et al. (2017) for more on how mean blood lead and ambient lead levels rapidly decreased following the enactment of leaded gasoline regulations in Korea and China.

[8] Workers at Du Pont's first TEL plant began dying from lead poisoning within a month of operation, and over 80 percent of workers at General Motors and Standard Oil's first TEL plant died or had severe lead poisoning (Kitman 2000).

Despite lead's clear physiological effects, it is difficult to recover causal estimates using observational data because exposure is not randomly assigned. In addition to gasoline and paint, lead exposure is a byproduct of a host of processes including refining, manufacturing, power generation, smelting, mining, and even drinking water transportation. Many of these processes generate substantial disamenities for nearby residents, including increased exposure to lead and other pollutants (e.g., Currie et al. 2015; Klemick, Mason, and Sullivan 2019).

Lead exposure is also correlated with a number of potentially confounding factors like year of birth, place of residence, and socioeconomic status. The types of places that have leaded pipes, high soil lead content, or are near polluting facilities are also different from locations with lower lead exposure in other ways. Leaded pipes and pipes that contain leaded solder are more likely to be found in older homes (Levin et al. 2008), and industrial facilities are more likely to be located in disadvantaged communities (Ash and Boyce 2018; Christensen, Sarmiento-Barbieri, and Timmins 2020). All else equal, low-income and minority communities face higher ambient pollution levels and greater exposure to lead (Levin et al. 2008, Miranda et al. 2011), and even within a given community, low-income residents may face greater lead exposure due to an increased prevalence of leaded paint (Jacobs et al. 2002). In general, socioeconomic status is tightly linked to both health and lead exposure (Meara 2001; Currie and Stabile 2003; Adams et al. 2004; Currie 2009; Cutler, Lleras-Muney, and Vogl 2012).

Recent work has taken significant care to address many of the potential confounders associated with nonrandom lead exposure. A common strategy is to use fixed effects estimators that exploit cohort or group level differences in exposure to lead. Previous work has used this strategy to show that lead exposure through drinking water has negative effects on cognition, fertility, and infant mortality (Ferrie, Rolf, and Troesken 2012; Clay, Troesken, and Haines 2014; Grossman and Slusky 2019), that the Flint lead crisis was capitalized into housing values (Christensen, Keiser, and Lade 2019), that early life interventions may offset some of the negative effects of lead exposure (Billings and Schnepel 2018, Sorensen et al. 2019), and that leaded aviation gasoline increases the blood lead levels of those who live near airports (Zahran et al. 2017).[9] Another common approach is to use a variable correlated with exposure to lead as an instrument, such as distance from a major road or state average gasoline lead content, in order to address issues with nonrandom lead exposure, avoidance behavior, or measurement error. This strand of literature has shown that lead can increase rates of crime, delinquent behavior, and enrollment in special education, as well as decrease fertility and test scores (Reyes 2007, 2011, 2015; Feigenbaum and Muller 2016; Gazze 2016; Aizer et al. 2018; Clay, Portnykh, and Severnini 2018; Aizer and Currie 2019; Clay, Portnykh, and Severnini 2019). These empirical approaches often exploit leaded gasoline phasedowns to generate variation

[9] In addition to the lead literature, this paper also complements the broader pollution literature on the impacts of vehicle emissions (Auhammer and Kellogg 2011; Currie and Walker 2011; Knittel, Miller, and Sanders 2016), the impacts of toxics (Currie et al. 2015, Persico and Venator 2018), and the effects of lead remediation (Billings and Schnepel 2017, Gazze 2020, Sorensen et al. 2019).

in lead exposure that is plausibly exogenous to socioeconomic confounders (Reyes 2007, 2011, 2015; Grönqvist et al. 2018; Clay, Portnykh, and Severnini 2018).

One challenge with using historic leaded gasoline reductions to generate variation in lead exposure is that these reductions did not occur in isolation. Legislation intended to reduce ambient concentrations of lead and other pollutants was often introduced at similar times. As an example, consider the case of the US lead phasedown that began in the 1970s following the passage of the Clean Air Act. The Clean Air Act established National Ambient Air Quality Standards (NAAQS) for six criteria pollutants, and a number of programs and regulations were implemented to maintain compliance with these standards. Aggregate emissions of both directly and indirectly targeted pollutants gradually declined by 73 percent from 1970 to 2016 as a result of this portfolio of programs (US Environmental Protection Agency 1989, 1990; Lee 1991; Greenstone 2002; Schmalensee and Stavins 2018).[10]

Some of these regulations, such as catalytic converter mandates, simultaneously targeted both lead and other pollutants. In the United States, the key driver of the leaded fuel emissions reductions was the mandated adoption of catalytic converters in new model-year cars beginning in 1975 (Newell and Rogers 2003). Catalytic converters are emission control devices that convert CO, nitrogen oxides ($NO_x$), and VOCs into benign compounds like water and nitrogen (Newell and Rogers 2003; Ravindra, Bencs, and Van Grieken 2004). The pollutants controlled by catalytic converters have been linked to adverse health outcomes by a large body of prior work, and catalytic converter usage has been linked to large improvements in health in the United States and other countries (Mott et al. 2002, Currie and Neidell 2005, Greenstone and Hanna 2014, Alexander and Schwandt 2019). Although catalytic converters do not directly reduce lead emissions, lead damages the devices, effectively prohibiting leaded fuel use. As new cars with catalytic converters replaced older cars, emissions of lead, CO, $NO_x$, and VOCs all declined simultaneously at the point source.

Jointly, the portfolio of pollution policies deployed in the United States and other countries led to a gradual and correlated decline in the ambient concentrations of many pollutants. This presents problems for separately identifying the effects of lead from the effects of other pollutants. Our research setting overcomes the challenges in separately identifying the effect of lead since race cars do not use catalytic converters or other emission control devices, and the removal of lead was the only relevant change made to the fuel between 2006 and 2007.

[10] Online Appendix Figure A1 shows the consistent declines in mean ambient concentrations of all pollutants except for ozone. This general pattern of correlated, overlapping pollution regulations exists in countries other than the United States. For example, Swedish air pollution regulations began in 1969 with the passage of the Swedish Environmental Protection Act. Following their EPA formation, many new laws were implemented to address various sources of emissions such as high-sulfur fuel oil, mercury, pesticides, and lead in gasoline. This generated simultaneous decreases in ambient concentrations of a wide range of pollutant (Lundgren 2010). Sweden also began targeting reductions in motor vehicle emissions in 1969 and tightened emissions restrictions several times thereafter. This occurred simultaneously with leaded gasoline phasedowns in 1972 and mandated adoptions of catalytic converters in 1988 (Lovei 1998). See Lundqvist (1979) for a brief policy overview and comparison to the United States.

## II. Data

Here we describe our data. Summary statistics can be found in Section A.4 of the online Appendix.

**Racing Data:** Race data were obtained from Racing Reference, a site that catalogs historical races for NASCAR, ARCA, and a number of other racing series. We collected data on the universe of races for the four racing series that were required to switch to unleaded fuel in 2007: NASCAR's three national series and the ARCA Menards Series. For robustness checks, we also collected race data for all regional NASCAR K&N Pro Series, which are the main series for which we cannot determine the leaded fuel status at any point in time in our sample. For each race we record the number of drivers, the number of laps needed for race completion, the length of each lap, the actual number of miles driven in the race by each driver, the racetrack location, and the date of the race. Race miles are observed and not estimated from lap length. We have information on every race in each of the national series since their inception: 1949 for the Monster Energy Cup, 1982 for the Xfinity Series, and 1995 for the Gander Outdoors Truck Series.[11] Track location was ascertained through Google Maps and manually confirmed. Online Appendix Figure A3 shows the geographic distribution of counties that have a racetrack or border a county with a racetrack.

For a placebo test, we also collected data on the NTT IndyCar Series races from ChampCarStats. IndyCar races are an ideal comparison group since they do not use leaded fuel and because they use many of the same tracks as NASCAR and ARCA, but take place at different times of the year.[12] IndyCar allows us to test whether or not our ambient lead estimates are simply picking up on lead emissions from spectators' vehicles and generators or resuspension of legacy lead.[13]

**Pollution Data:** Daily pollution data were obtained from the EPA's AirData datasets, which report pregenerated daily summaries by pollutant at each air enforcement monitor. For every recorded monitor day from 1980 to 2018, data were collected for CO, $NO_2$, $PM_{10}$, $PM_{2.5}$, Pb, VOCs, and $O_3$. Daily data on lead levels prior to 1980

[11] These series have gone by several different names since their inception. We use their 2019 names when discussing them in this paper.

[12] IndyCar has not used lead additives since 1964 when it began using pure methanol, an alcohol-based fuel. In 2006, it switched to a 90/10 ethanol/methanol blend, and in 2007 it switched to a 98/2 ethanol/gasoline blend.

[13] In addition to NASCAR, ARCA, and IndyCar, there are other racing series in the US such as Formula 1, National Hot Rod Association (NHRA) drag racing, and a large number of regional and local series. Like NASCAR, Formula 1 historically allowed leaded fuel but switched to unleaded gasoline in 1996 (Collins 2012). We omit Formula 1 races from our analysis because there is at most one Formula 1 race per year in the United States, and the fuel switch occurred prior to the earliest years used in our results. We do not include data on NHRA series since racers have the option to use a wide range of fuels including gasoline, diesel, propane, or nitromethane. In the series that require a specific fuel (e.g., the Mello Yello Drag Racing Series, which requires either nitromethane or methanol) we are unaware of any changes to fuel rules that coincide with NASCAR and ARCA deleading. In Section A.5 of the online Appendix, we discuss changes in car performance across time, where racetracks are located, where lead monitors are located relative to racetracks, and other events that occur at the racetracks in more detail.

were provided by the EPA by request. All events were included in the calculation of a daily mean.

**Weather Data:** Weather data come from the National Oceanic and Atmospheric Administration's (NOAA) NCEP-NCAR Reanalysis 1 model, which provides daily weather data in a 2.5° latitude by 2.5° longitude global grid (Kalnay et al. 1996). We extracted data from 1948 to 2018 on surface level (i.e., 0.995 sigma level) average air temperature (C°), pressure (kPA), relative humidity (percent), and wind. Wind data are reported within each cell by two vectors: a u-wind vector that measures east to west wind in meters per second and a v-wind vector that reports north to south wind in meters per second. Using these two vectors, we calculate wind speed $(\mathrm{m}/\mathrm{s})$ and wind direction.[14] To proxy for precipitation, we extracted data on daily precipitable water $(kg/m^2)$ in each 2.5° by 2.5° atmospheric column; these data are not reported at the surface level. To estimate the daily weather conditions at each pollution monitor, we first determine the 2.5° latitude by 2.5° longitude grid cell in which each pollution monitor is located. We then assign the daily weather data for each cell to all pollution monitors within it.

**Blood Lead Data:** Blood lead data were obtained from the Centers for Disease Control and Prevention (CDC) State Surveillance Data. Data are reported at the county-year level from 2005 to 2015. The blood lead data report the percentage of children tested in a given county-year that had elevated blood lead levels, which would be a BLL above 10 $\mu\mathrm{g}/\mathrm{dl}$. The CDC funded 29 states, Washington, DC, and 5 cities to sample blood lead in children under 72 months old. Five additional states also voluntarily submitted data. Blood lead sampling was targeted at high-risk areas. Online Appendix Section A.7 describes two tests of whether targeting is a concern for identification. We do not find evidence that this is the case.

Here we focus on confirmed cases of elevated blood lead. Confirmed cases of elevated blood lead in children require a venous blood draw registering a BLL above 10 $\mu\mathrm{g}/\mathrm{dl}$, or two separate capillary draws within two weeks registering BLLs above 10 $\mu\mathrm{g}/\mathrm{dl}$. This helps avoid measurement error in readings from a single capillary draw, which can be easily contaminated.

**Mortality Data:** Mortality data come from the restricted CDC National Vital Statistics System, and population data come from the National Cancer Institute's Surveillance, Epidemiology, and End Result (SEER) program. Mortality data report information at the individual level of death. We observe cause of death; age at death; county, month, and year of death; and county of residence. The SEER data are reported from 1999 to 2016. We report the age-standardized death rate per 100,000 population of each cohort. The reference population we use to standardize is the US population for each cohort in 2000. For our analysis, we focus on the

[14] Wind speed $= \sqrt{u^2+v^2}$. Wind direction $= \mathrm{mod}\big(\arctan2(-u,-v)\cdot(180/\pi)-180,360\big)$.

elderly population (65+), but we also report results for infants (under 1 year old) in Section A.8 of the online Appendix.

**Covariate Data:** In some specifications, we control for socioeconomic variables.[15] We collect county-level data on median income from the US Census, unemployment rates from the Bureau of Labor Statistics, and percent non-white from the SEER program. We obtain data from the Toxic Release Inventory on lead emissions from industrial facilities. Facilities included in the TRI emissions data must have at least 10 employees and manufacture, process, or use above 100 pounds of general lead compounds. The threshold is 25,000 pounds if the facility is using lead in brass, stainless steel, or bronze alloys. We augment the lead emissions data with county level payroll data for the manufacturing sector from the County Business Patterns dataset as a way to proxy for other emissions, address the censoring of smaller facilities, and address the known measurement error issues in the TRI database (de Marchi and Hamilton 2006).

## III. Methods and Results

In this section, we describe our empirical methods and present our results for each outcome of interest. All of our results exploit the deleading of NASCAR and ARCA fuel in 2007. Due to differences in the granularity of our outcomes data across both time and space, we use a slightly different specification for ambient lead outcomes versus the blood lead and mortality outcomes. In addition, we choose geographic units of comparable size across each analysis, further ensuring that results are commensurable. Sections A.6–A.9 in the online Appendix contain alternative specifications, robustness checks, and additional analyses for all outcomes.

### A. *Ambient Lead Concentrations*

*Ambient Lead Concentrations Methods*.—To estimate the effect of leaded races on ambient lead concentrations we estimate an event study and a spatial lag model. The following equation is our simplest specification, a monitor-day design that underlies results displayed in the tables:

$$\text{(1)}\quad \text{arcsinh}(Pb)_{mdwy} = \beta_l \, leaded\ race\ miles_{mdwy} + \beta_u \, unleaded\ race\ miles_{mdwy} + \gamma \mathbf{X}_{mdwy} + \Theta_{my} + \Omega_{wy} + \varepsilon_{mdwy}.$$

The function $\text{arcsinh}\left(Pb\right)_{mdwy}$ is the inverse hyperbolic sine of the measured lead concentration at monitor $m$ on day $d$ in week-of-year $w$ and year $y$. We use the arcsinh transform because the empirical lead concentration distribution is right-skewed

[15] In Section A.9.ii of the online Appendix, we report how counties with racetracks differ from other counties along socioeconomic indicators that were omitted from our set of covariates such as percent in poverty. We find that conditional on covariates and fixed effects, race counties, border counties, and control counties are similar on these margins.

and has a large number of zero-valued observations. The arcsinh transform is similar to the natural log transform insofar as it is monotonic and compresses the influence of large outliers. It does this with the additional advantage of preserving zero-valued inputs without needing additional manipulation.

The term $\mathbf{X}_{mdwy}$ is a vector of covariates containing the average temperature, precipitable water, relative humidity, surface pressure, and wind speed on the day of the reading; $\Theta_{my}$ is a set of monitor-by-year fixed effects that flexibly control for monitor-specific annual trends in lead concentrations, and $\Omega_{wy}$ is a set of week-by-year fixed effects that control for within-week shocks that are common across monitors; $\varepsilon_{mdwy}$ is the error term. Standard errors are robust to heteroskedasticity and clustered at the county level since the errors of nearby monitors are likely to be correlated. In our main results, we restrict our sample to begin in 1996, when on-road leaded fuel was banned. This avoids potential confounding from spectator travel to the racetrack.

The variable *leaded race miles*$_{mdwy}$ is the number of leaded race miles that occurred within 50 miles of monitor $m$, in the week prior to monitor $m$'s lead reading on day $d$.[16] The variable *unleaded race miles*$_{mdwy}$ is the same but for unleaded race miles. Our coefficients of interest are $\beta_l$ and $\beta_u$; $\beta_l$ is the effect of the leaded race miles driven in the past week on ambient lead readings within 50 miles, and $\beta_u$ is the same but for unleaded miles. Identification of $\beta_l$ comes from variation in leaded miles at a monitor, within a year, off of week-of-year averages across monitors.[17] Identification of $\beta_u$ comes from the same variation but in unleaded race miles.

To estimate a distance gradient we use a spatial lag model. This specification also includes four variables in addition to those in equation (1). The four variables are the number of leaded and unleaded miles that occurred within the last week between 50–75 miles and 75–100 miles away from the monitor. To estimate a time gradient and test for any pre-trends in lead concentrations prior to a race, we use an event study specification. This specification includes 14 additional variables: the number of leaded and unleaded race miles that occurred within 50 miles, but 8–14 days, 15–21 days, and 22–28 days in the past; and the number of leaded and unleaded race miles that occurred within 50 miles, but 0–7 days, 8–14 days, 15–21 days, and 22–28 days in the future.

*Ambient Lead Concentrations Results*.—First, we plot the raw data. Figure 2 plots the average ambient concentration of lead by weeks since the most recent race in the 11 years prior to deleading (panel A) and the 11 years after deleading (panel B). Negative values on the $x$-axis indicate monitor readings that occurred in the weeks prior to a race, and positive values on the $x$-axis indicate monitor readings that occurred in the weeks after a race. During leaded race years, lead concentrations prior to a race show a slight downward trend within 50 miles of a racetrack and are flat or slightly increasing further away. In the week after a race, concentrations jump

[16] Section A.6 of the online Appendix shows our results are robust to using alternative cutoffs to 50 miles.

[17] Note that identification of $\beta_l$ does not rely upon unleaded race miles being driven. We are comparing monitor readings on days with positive leaded race miles in the past week to monitor readings on days without. These are primarily days without any kind of race in the past week. The same is true for identification of $\beta_u$ and leaded miles.

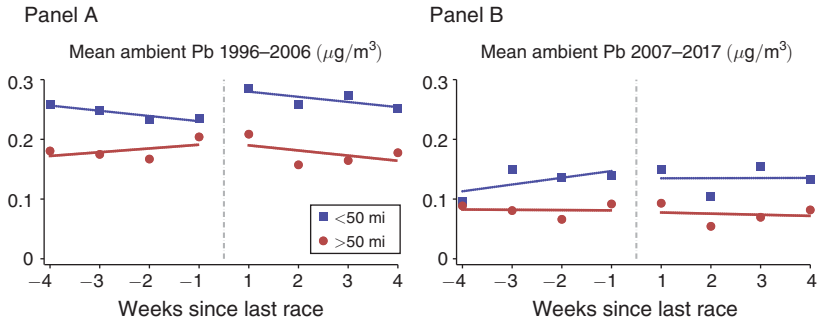



Figure 2. Ambient Lead Concentrations in the Four Weeks before and after the Last Leaded Race (Left) or Unleaded Race (Right)

*Notes:* This figure plots ambient lead concentrations across time and space. Panel A plots average ambient lead concentrations four weeks before and after a race from 1996 to 2006, the years when races were leaded. Panel B is the same, but for 2007–2017, the years when races were unleaded. Negative values on the *x*-axis indicate monitor readings that occurred in the weeks prior to a race, and positive values on the *x*-axis indicate monitor readings that occurred in the weeks after a race.

by over 20 percent compared to concentrations the week prior at monitors within 50 miles. The jump slowly attenuates until it reaches pre-race levels after 4 weeks. There is essentially no change in concentrations outside 50 miles. Panel B shows that once NASCAR and ARCA deleaded, the post-race jump in ambient lead concentrations vanishes.

Table 1 displays estimates that gradually build up to our preferred set of fixed effects and controls in column 7, which corresponds to equation (1).[18] The top row shows estimates of the effect of 100,000 leaded miles in the past week, and the second row shows estimates of the effect of 100,000 unleaded miles in the past week. Column 1 controls for only monitor fixed effects and year fixed effects. In this simple comparison, the estimated effect of 100,000 leaded miles on arcsinh(Pb) is 0.16, and the effect of 100,000 unleaded miles is 0.07. As we add monitor-specific daily weather controls (column 2) and week-of-year fixed effects (column 3), neither coefficient meaningfully changes. Including state-by-year fixed effects (column 4) reduces the impact of leaded miles to 0.13 and the impact of unleaded miles to 0.02. Further refining the spatial fixed effects to county-by-year (column 5) or monitor-by-year (column 6) and adding week-by-year fixed effects (column 7) continues to attenuate the estimated relationship between unleaded race miles and ambient lead levels, while leaving the estimated effect of leaded race miles on ambient lead concentrations relatively unchanged.

The inclusion of flexible time-trends (at the state, county, or monitor level) are important for our identification since they help guard against the possibilities that our estimates only reflect spurious differential trends (e.g., declining ambient lead levels in areas far from racetracks) or that our findings are caused by local changes in pollution, policy, or other unobservables (e.g., the closing of a manufacturing plant). Incorporating monitor-by-year and week-by-year fixed effects, as we do in

[18] In the sample there are 205 lead monitors that were treated at least once by a leaded race within 50 miles, and 643 lead monitors that had never been treated.

Table 1—The Effect of Race Miles on Ambient Lead Concentrations at Monitors within 50 Miles of a Racetrack

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Leaded race miles in past week (100k) | 0.16 | 0.14 | 0.14 | 0.13 | 0.13 | 0.13 | 0.13 |
| | (0.08) | (0.08) | (0.07) | (0.06) | (0.05) | (0.05) | (0.05) |
| Unleaded race miles in past week (100k) | 0.07 | 0.06 | 0.06 | 0.02 | 0.01 | 0.01 | –0.00 |
| | (0.03) | (0.02) | (0.02) | (0.01) | (0.01) | (0.01) | (0.01) |
| Monitor fixed effects | Yes | Yes | Yes | Yes | Yes | No | No |
| Year fixed effects | Yes | Yes | Yes | No | No | No | No |
| Daily weather controls | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Week fixed effects | No | No | Yes | Yes | Yes | Yes | No |
| State-by-year fixed effects | No | No | No | Yes | No | No | No |
| County-by-year fixed effects | No | No | No | No | Yes | No | No |
| Monitor-by-year fixed effects | No | No | No | No | No | Yes | Yes |
| Week-by-year fixed effects | No | No | No | No | No | No | Yes |
| Clustered standard errors | County | County | County | County | County | County | County |
| Adjusted $R^2$ | 0.35 | 0.35 | 0.35 | 0.39 | 0.41 | 0.44 | 0.45 |
| Observations | 312,296 | 312,296 | 312,296 | 312,291 | 312,285 | 312,277 | 312,277 |

*Notes:* Robust standard errors clustered at the county level are in parentheses. The dependent variable is arcsinh(Pb). Monitor-specific daily weather variables include air temperature, pressure, relative humidity, wind speed, and daily precipitable water. Column 7 shows estimates from equation (1).

our preferred specification in column 7, ensures that our treatment effect estimates are derived from within monitor-year variation in the quantities of leaded and unleaded miles, and address common seasonal patterns in ambient concentrations. Using this specification, we estimate that every 100,000 leaded miles driven in the past week increases ambient arcsinh(Pb) levels by 0.13, which is equivalent to a 10 percent increase in mean concentrations.[19] We further estimate a precise zero effect of unleaded miles on ambient lead concentrations; we can statistically distinguish the leaded and unleaded miles estimates from each other.

Figure 3 plots the estimated time and distance gradients. Panel A plots our event study estimates of the effect of 100,000 race miles by weeks since the last race within 50 miles. In the weeks prior to exposure to either type of race miles, we do not detect any change in ambient lead concentrations. In the first week after exposure to 100,000 leaded miles, we estimate that arcsinh(Pb) increases by 0.15, an 11 percent increase in mean concentrations. The effect then tapers off toward zero in the weeks after. All estimates for unleaded miles are centered around zero with no clear trend over time, suggesting that unleaded races do not resuspend legacy lead or have other attributes that increase ambient lead levels.

Panel B of Figure 3 plots spatial lag estimates of the effect of 100,000 race miles in the past week on arcsinh(Pb) by distance bands. We find an intuitive decreasing gradient in distance for leaded miles. We estimate an effect of 10 percent in the 0–50-mile distance band, which declines to 2 percent in the 50–75-mile band, and to approximately zero in the 75–100-mile band. In contrast, we do not find any discernible relationship between unleaded miles and distance from racetrack to monitor.

[19] See Bellemare and Wichman (2019) for how to compute arcsinh elasticities. They are slightly different than log elasticities, so the percentage changes we report will not always match the coefficient estimate.

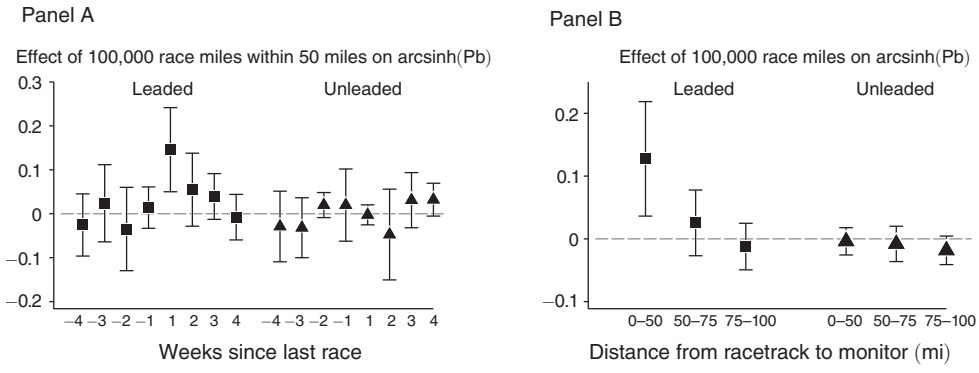



FIGURE 3. THE EFFECT OF RACE MILES ON AMBIENT LEAD BY WEEKS SINCE MOST RECENT RACE (LEFT) AND BY DISTANCE FROM TRACK (RIGHT)

*Notes:* This figure shows the effect of 100,000 leaded and unleaded race miles on ambient lead concentrations across both time and space. Panel A plots the estimated time gradient, while panel B plots the estimated distance gradient. For each panel, all coefficients come from one regression. Each regression controls for week-by-year fixed effects, monitor-by-year fixed effects, and daily monitor-specific weather controls. The dependent variable is arcsinh(Pb). Brackets denote 95 percent confidence intervals, calculated from robust standard errors clustered at the county level. For the time panel (A), negative values on the *x*-axis indicate monitor readings that occurred in the weeks prior to a race, and positive values on the *x*-axis indicate monitor readings that occurred in the weeks after a race. For the time panel, only race miles within 50 miles are included. For the distance panel (B), only race miles within the past seven days are considered.

**Robustness Checks and Alternative Specifications:** Section A.6 of the online Appendix further demonstrates the robustness of our estimates to a variety of specifications and sample definitions. It also contains event study estimates including baseball and IndyCar as additional comparison groups showing they have no effect on ambient lead concentrations, and that there is no differential effect of leaded versus unleaded races on concentrations of other potentially confounding pollutants. Section A.9.i contains a cross-sectional analysis of soil lead, showing higher concentrations near racetracks. In online Appendix Section A.10, we use our estimates to predict how daily historic vehicle traffic affects mean air quality in major cities. Over the range of common support, our estimates predict historic mean lead levels quite well, lending further credibility to our in-sample estimates. This section also reports results from a counterfactual prediction of ambient lead concentrations as if lead were still added to gasoline at 1970 levels.

## B. *Blood Lead*

*Blood Lead Methods.*—Our elevated blood lead level outcomes are at the county-year level, so we use the following event study specification:

$$\begin{aligned}(2)\qquad \operatorname{arcsinh}(EBLL)_{scy} = &\sum_{t\in\mathcal{Y}} \beta_t^r \mathbf{1}(\textit{race in county})_{sct} \\ &+ \sum_{t\in\mathcal{Y}} \beta_t^b \mathbf{1}(\textit{race in border county})_{sct} \\ &+ \gamma \mathbf{X}_{scy} + \Theta_c + \Omega_{sy} + \varepsilon_{scy}.\end{aligned}$$

Here, $\text{arcsinh}(EBLL)_{scy}$ is the inverse hyperbolic sine of the elevated blood level prevalence in state $s$, county $c$, and year $y$. We again use the inverse hyperbolic sine transform because of the distribution of elevated blood lead rates is right-skewed and contains a large number of zero-valued observations. The term $\sum_{t\in\mathcal{Y}}\mathbf{1}(race\ in\ county)_{sct}$ is our main set of treatment variables. They are indicator variables that are equal to 1 if there was a race in county $c$ in year $t$.[20] The term $\mathcal{Y}$ is the set of years from 2005 to 2015, with 2007 omitted. The $\beta_t^r$ terms capture the effect of having a race in a year $t$ on $\text{arcsinh}(EBLL)_{scy}$ in year $y$, relative to the difference in $\text{arcsinh}(EBLL)_{scy}$ between race counties and control counties in 2007. Control counties in year $t$ are those that did not have a race and did not border a county with a race in year $t$. This does allow for some counties to be in the control group in one year and in one of the treatment groups in another. However, since the half-life of blood lead is short, and there are generally not NASCAR or ARCA races in the first or last 40 days of the year, effects of leaded races on the prevalence of elevated blood lead in the next year would be small.

The term $\sum_{t\in\mathcal{Y}}\mathbf{1}(race\ in\ border\ county)_{sct}$ is a set of indicator variables equal to 1 if there was a race in a county that borders county $c$ in year $t$, but there was no race in county $c$ itself. The $\beta_t^b$ terms capture the effect of not having a race, but bordering a county with a race in a particular year on $\text{arcsinh}(EBLL)_{scy}$, relative to the difference in $\text{arcsinh}(EBLL)_{scy}$ between border counties and control counties in 2007. This second set of variables is to determine if there is a distance gradient for EBLL rates like we found for ambient lead.

The vector $\mathbf{X}_{scy}$ is a set of controls that includes the unemployment rate, median income, percent non-white, payroll in the manufacturing sector, and quantity of TRI lead emissions. The first three are to control for socioeconomic status and potential sorting across locations, while the latter two are to control for alternative sources of lead emissions.[21] The variable $\Theta_c$ is a set of county fixed effects, and $\Omega_{sy}$ is a set of state-by-year fixed effects; $\varepsilon_{scy}$ is the error term. Standard errors are robust to heteroskedasticity and are clustered at the county level. Identification of $\beta_t^r$ and $\beta_t^b$ comes from annual within-state variation in races, off of the county means. Since the sample of children tested is a nonrandom sample of those at risk for higher blood lead levels, an additional identification assumption is that this selection did not differentially change for race and border counties relative to control counties around the time of deleading. This is addressed further in online Appendix Section A.7, where we show that the number of children tested in race counties did not change in race counties following deleading.

[20] The sum of leaded and unleaded race miles in a county does not change much from year to year since the race schedule is usually the same on an annual basis, and races typically maintain the same length over time. The only variation in the sum of the two types of miles generally comes from weather-related issues, or crashes or retirements during a race. Since total race miles is approximately fixed and total race miles = leaded miles + unleaded miles, using total miles in a given county-year as treatment variables will be effectively the same as using dummy variables after conditioning on county fixed effects, and trying to separately estimate the effect of leaded and unleaded miles will have the usual multicollinearity issues. Figure A20 of the online Appendix shows the high degree of collinearity between leaded and unleaded miles. We explore outcomes in a difference-in-differences framework with varying treatment intensity using just leaded miles in Section A.7 of the online Appendix.

[21] Section A.9.v. of the online Appendix contains results from a Mechanical Turk survey indicating that the public may be generally unaware of the fuel used by NASCAR and when they have switched fuels. This alleviates concerns about sorting and other behavior that may bias our estimates.

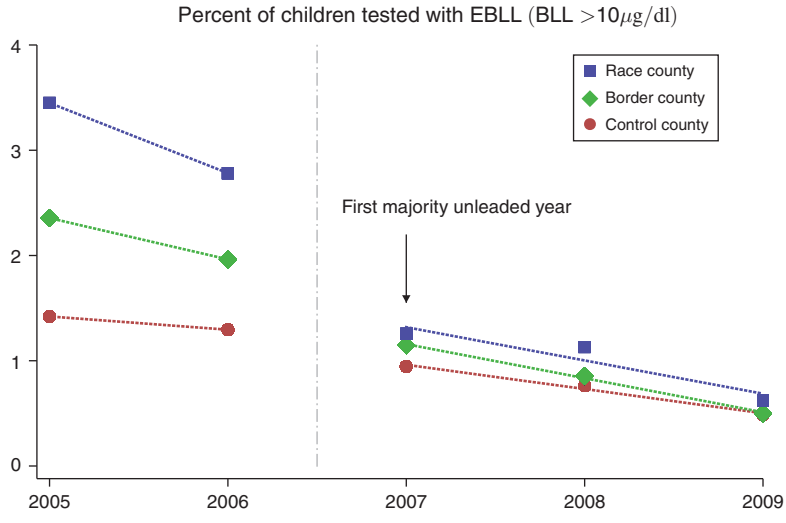



FIGURE 4. MEAN PERCENT OF CHILDREN WITH ELEVATED BLOOD LEAD LEVELS BY YEAR AND COUNTY TYPE

*Notes:* Each point represents the mean percent of children tested with elevated blood lead levels by county type, weighted by the square root of the number of children tested. The results are similar when we do not weight the data. Elevated blood lead level means that the concentration of lead in the blood is at least 10 $\mu$g/dl. County type refers to if there was a NASCAR or ARCA race in that county (blue squares) or in a border county (green diamond) in that year. All other counties are considered control counties (red circle). For this figure we use a balanced panel of counties from 2005 to 2009. All data are included in our regression estimates.

*Blood Lead Results.*—We begin by plotting the raw data. Figure 4 plots average prevalence of elevated blood lead from 2005 to 2009 by county type, where the means are weighted by the square root of the number of children tested.[22] Prior to the switch to unleaded fuel in 2007, all three types of counties exhibited slight downward trends in EBLL rates that are nearly parallel, although the rate of elevated blood lead in border counties and race counties were 1 and 2 percentage points higher than in control counties. In 2007, when NASCAR and ARCA permanently switched to unleaded gasoline, control county EBLL rates continued along the same trend, but race and border county EBLL rates dropped so that they were nearly identical to control counties. In subsequent years, EBLL rates across county types had similar levels and parallel trends.

Figure 5 plots estimates from our main regression. The top panel displays estimates for race counties. In 2005 and 2006 we estimate that race counties had EBLL rates that were 18 percent and 13 percent higher than in 2007. The decrease from 2005 to 2006 is consistent with the fact that 14 percent of race miles driven in 2006 were unleaded race miles, thus decreasing exposure in race counties relative to 2005. For all years after 2007, the point estimates are all close to zero. The dashed line is the point estimate from our preferred difference-in-difference regression, where year-specific race county and border county indicator variables are replaced with

[22] Leaving the data unweighted does not affect the results. We truncate the sample at 2009 for the plot since after 2009 some counties stop reporting and the composition of our sample changes.

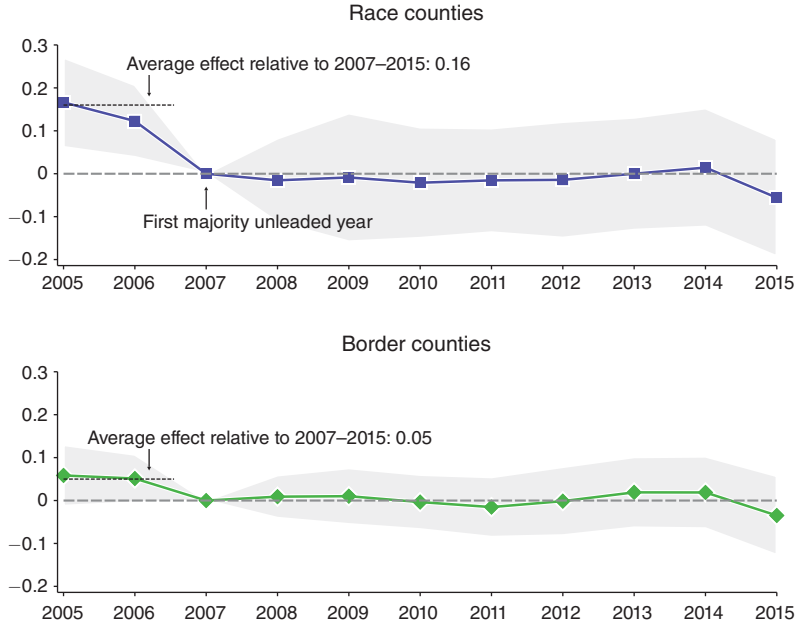



FIGURE 5. THE EFFECT OF LEADED RACES ON THE PERCENT OF CHILDREN WITH ELEVATED BLOOD LEAD LEVELS IN RACE AND BORDER COUNTIES

*Notes:* The top panel reports coefficients (blue squares) for children living in race counties. The bottom panel reports coefficients (green diamonds) for children living in border counties. Each coefficient represents the effect of being in a particular county type relative to 2007, which is omitted. The regression includes state-by-year fixed effects, county fixed effects, and controls for the unemployment rate, median income, percent non-white, tons of lead emitted from TRI facilities, and total manufacturing payroll. All coefficients come from the same regression. The regression is weighted by the square root of the number of children tested. The shaded gray areas denote the 95 percent confidence interval calculated from robust standard errors clustered at the county level. The dashed line is the average effect from our preferred difference-in-differences regression, where the race county treated group consists of those counties that had at least one leaded race prior to 2007, and the border county treated group consists of those counties that did not have a leaded race but bordered a county with a leaded race prior to 2007. This regression defines the post-period as 2007 and after.

two indicator variables: one for if there was a race in the county in 2005 or 2006, and one for if there was not a race in the county but there was one in a bordering county in 2005 or 2006. The estimates from this regression indicate that the prevalence of elevated blood lead was on average 17 percent higher in leaded race years versus unleaded race years.

The bottom panel shows estimates for border counties. Here we estimate an effect of 6 percent in 2005 and 5 percent in 2006 relative to 2007, and effects that are close to zero in the years after 2007. The difference-in-difference regression yields an estimate of 6 percent higher EBLL rates in leaded races years versus unleaded race years. The smaller estimated effect for border counties is consistent with a decreasing distance gradient. Section A.7 of the online Appendix includes estimates from the difference-in-difference regressions as we build up to our main specification, event study estimates when the outcome is the untransformed EBLL rate, and additional estimates using the quantity of leaded miles as treatment.

Our event study results shed light on the mechanisms driving higher EBLL rates in race counties. EBLLs could be caused by exposure to newly emitted lead from

races, exposure to legacy lead deposits, or some combination of the two. The switch to unleaded fuel affects the flow of new lead, which reduces airborne exposure and additional deposition to the existing stock of legacy lead. If the effects on EBLL rates were mainly a function of the stock of lead, we should expect the decrease in EBLL rates in race counties relative to control counties to occur more slowly over many years or perhaps to not find a decline at all. The observed, immediate decline in 2007 and subsequent plateau in both the regression estimates and raw data is consistent with higher EBLL rates being primarily a function of exposure to lead emissions from recent races.

Online Appendix Section A.9.ii reports results from a series of specifications that examine if omitted variables or differential trends could be driving the estimates between leaded races and the prevalence of elevated blood lead reported above. We follow Pei, Pischke, and Schwandt (2019) and swap our dependent variable of interest with one of a number of socioeconomic variables not included as control variables in our preferred specification. We find no clear, statistically significant, or economically meaningful relationship between our independent variables of interest and these socioeconomic variables, helping to alleviate concerns that our estimates are driven by omitted variables or differential trends.

## C. *Elderly Mortality*

*Elderly Mortality Methods.*—Here we use the same empirical approach as the EBLL analysis since the data are at the same county-year level. The specification is

$$\begin{aligned} \textit{mortality rate}_{scy} = & \sum_{t \in \mathcal{Y}} \beta_t \mathbf{1}(\textit{race in county})_{sct} + \sum_{t \in \mathcal{Y}} \beta_t^b \mathbf{1}(\textit{race in border county})_{sct} \\ & + \gamma \mathbf{X}_{scy} + \Theta_c + \Omega_{sy} + \varepsilon_{scy}. \end{aligned}$$

The variable $\textit{mortality rate}_{scy}$ is the cause-specific, age-standardized, elderly mortality rate in state $s$, county $c$, and year $y$. Since the mortality rate distributions are roughly normally distributed with very few zeros we do not use the inverse hyperbolic sine transformation.[23] The mortality rate is in terms of deaths per 100,000 in the elderly cohort, which we define as those aged 65 and above. The rest of the specification is identical to equation (2). We study six causes of death. The first is all-cause mortality. This picks up an aggregate mortality impact and avoids potential misclassifications related to any particular cause of death. The next four causes of death are cardiovascular mortality, ischemic heart disease (IHD), respiratory mortality, and deaths of despair as defined by Case and Deaton (2015). We selected cardiovascular, IHD, and respiratory mortality because of the existing associational evidence that EBLLs are related to cardiovascular mortality and reduced respiratory function (Pirkle et al. 1985, Lustberg and Silbergeld 2002, Jurdziak et al. 2015, Lanphear et al. 2018). We included deaths of despair because higher levels of blood or bone lead are associated with higher rates of drug addition, psychiatric symptoms

[23] Results are similar if we use an inverse hyperbolic sine transformation.

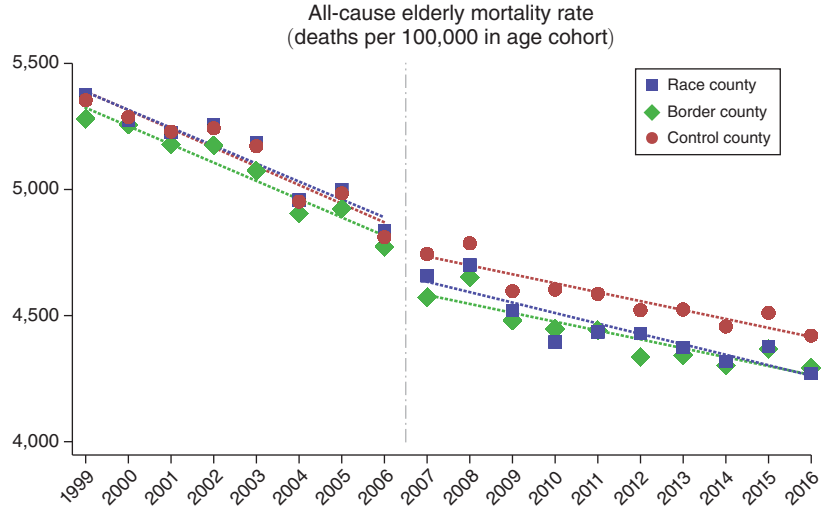



FIGURE 6. MEAN ALL-CAUSE ELDERLY MORTALITY RATES BY YEAR AND COUNTY TYPE

*Notes:* Each point represents the mean age-standardized all-cause elderly mortality rate by county type weighted by the square root of elderly population. The elderly population of the entire United States in the year 2000 was used as the reference population for standardization. County type refers to if there was a NASCAR or ARCA race in that county (blue squares), or in a border county (green diamonds) in that year. All other counties are considered control counties (red circles). For this figure and in our regression estimates we use a balanced panel of counties.

such as depression and panic attacks, and suicide risk (Rhodes et al. 2003, Fishbein et al. 2008, Bouchard et al. 2009, Kim et al. 2015).

The last cause of death is a placebo: diabetes. Lead has a systemic effect on the human body making it difficult to come up with valid placebos a priori. However, since diabetes is a chronic disease, the sudden deleading of gasoline is unlikely to affect diabetes mortality rates. We find no effect of deleading on diabetes; the event study estimates can be found in Section A.8.i of the online Appendix. Section A.8.i of the online Appendix also contains the specific CDC and International Classification of Diseases (ICD) codes used to define each of the six causes of death.

*Elderly Mortality Results.*—We again begin with the raw data. Figure 6 plots the raw data for the age-standardized average all-cause mortality rate, where counties are weighted by the square root of the elderly population. All three types of counties exhibited steep downward trends over the sample period. All-cause mortality rates were nearly identical in race and control counties prior to the fuel switch, while border county rates were slightly lower but follow a parallel trend. At the time of deleading in 2007, mortality rates of race counties dropped below those of control counties, and border county mortality rates also dropped relative to control counties. Trends in the post-period were also approximately parallel. Figure A14 in the online Appendix plots the data for cause-specific mortality and the figure shows similar patterns as for all-cause mortality.

Panel A of Figure 7 plots our all-cause mortality estimates in event study form. There are no clear pre-trends in the estimated effect of being in a race or border

county relative to a control county. All-cause mortality rates in race and border counties declined relative to control counties after deleading, and remained lower for the rest of the sample period. To get a better sense of the average size of the effect, the dashed line shows the estimated effect when we replace the event study with our preferred difference-in-difference specification with unleaded treatment beginning in 2007. We estimate a decline in elderly mortality rates of 91 deaths per 100,000 in race counties and 38 deaths per 100,000 in border counties.

Figure 7 also plots event study estimates for the other four mortality outcomes. Again, there are no clear pre-trends except for respiratory mortality. Mortality rates in race and border counties decrease relative to control counties at the time of deleading for all three causes of death, and then they remain lower. Here we find that cardiovascular mortality rates drop by 37 deaths per 100,000 in race counties after deleading, and drop by 12 deaths per 100,000 in border counties after deleading. IHD mortality rates drop by 53 deaths per 100,000 in race counties and 20 deaths per 100,000 in border counties; respiratory mortality rates drop by 20 deaths per 100,000 and 5 deaths per 100,000; and deaths of despair mortality rates drop by 4 deaths per 100,000 and 2 deaths per 100,000. In total, our results suggest that lead emissions have an immediate effect on elderly mortality. The smaller effects in border counties are consistent with the expected distance gradient.[24]

The online Appendix shows that we find weak evidence for effects on infant mortality, although an existing literature finds that lead exposure has negative effects on fertility, infant mortality, and infant health (Troesken 2008; Clay, Troesken, and Haines 2014; Grossman and Slusky 2019). There are several possible reasons for this distinction. First, our findings indicate that leaded gasoline tends to increase elderly mortality through cardiovascular causes and deaths of despair. The former requires long-term damage to the cardiovascular system while the latter is an adult cause of death. Second, our results are from a different medium of exposure. In our case, people are exposed to lead through the air while the previous literature has explored exposure through drinking water. Third, the level of exposure likely differs between our setting and others. We study a setting where ambient lead concentrations, as well as concentrations of other pollutants, are very low. If infant mortality increases only at higher levels of lead exposure, then we will not estimate an economically significant effect.

## IV. The Social Cost of Leaded Gasoline

We value the external effects of leaded gasoline in two ways. First, we estimate the effect on future earnings from increases in blood lead concentrations and the associated decreases in IQ. Second, we value the effect of lead on premature all-cause elderly mortality and use this to provide a lower bound on the total social cost of leaded gasoline.

[24] One potential explanation for our findings is that they are actually caused by the Great Recession. We argue that this is unlikely. If improvements in mortality rates were driven by differential effects of worsening economic conditions, we would expect the differential decreases in mortality in race and border counties to vanish as the recession ended and the economy improved. We find little evidence that mortality rates rebounded along with economic conditions.

### A. *Blood Lead, IQ, and Future Income*

To estimate the expected decreases in IQ from lead exposure, we combine estimates from our work with those from the prior literature. First, we translate our county level difference-in-difference estimates reported in Figure 5 into expected changes in individual BLLs. Recall that our county level estimates indicate that living in a race county increases the prevalence of those with EBLLs by 17 percent and that the threshold for EBLL is 10 $\mu$g/dl. First, we assume the 17 percent estimate is the probability that living in a county with leaded races causes a child to have an elevated BLL value. Second, we conservatively assume that if a child does not test positively for EBLL, their baseline BLL level is just below the threshold at 9 $\mu$g/dl, and if the child tests positively, their elevated BLL is right on the EBLL threshold at 10 $\mu$g/dl. We make this last assumption because we are interested in the minimum individual changes that are consistent with our observed estimates. This helps to provide as conservative a valuation as possible because it is unlikely that all blood lead levels below the elevated threshold dropped to zero following deleading. This procedure suggests that the deleading of NASCAR and ARCA races resulted in a $0.17 \times (10\ \mu\text{g/dl} - 9\ \mu\text{g/dl}) = 0.17\ \mu\text{g/dl}$ expected decline in BLLs per child in race counties.

Next, we link changes in BLLs to changes in IQ. The average IQ point loss per 1 $\mu$g/dl for BLLs under 10 $\mu$g/dl is 0.513 (Lanphear et al. 2005, Gould 2009).[25] In expectation, a child in a race county experienced a 0.087 IQ point decrease. Last, we link IQ to earnings following the procedure in Klemick, Mason, and Sullivan (2019), which estimates that a 1 IQ point decrease for a three-year-old is associated with a present value earnings loss of $20,568 in 2019 dollars.[26] Combining these estimates, the costs of reduced IQ are $1,789 per child in race counties.

Because blood lead testing is targeted at high-risk areas, it is unclear whether we can map this number to an aggregate county-wide benefit from deleading. Instead, we consider the deleading benefits only obtained by the children in the EBLL dataset. The dataset reports that 75,733 children under 6 were tested in race counties in 2015.[27] This gives us that the IQ-related social benefits for the cohort of children tested in 2015, who were never exposed to leaded NASCAR or ARCA races during their lifetime, is $135 million. When we perform the same exercise for border counties, we get that expected BLL declines are 0.06 $\mu$g/dl, and expected IQ increases are 0.031 points. This gives a present value earnings gain per child of $638 in 2019 dollars. In 2015, 242,385 children were tested in border counties for a total earnings gain of $155 million. In total, we estimate that the 2015 cohort of tested children in race and border counties obtained $290 million in benefits from never being exposed to leaded races.[28]

[25] Previous estimates suggest that the relationship may be roughly half this size: between 0.185 and 0.323 IQ points per 1 $\mu$g/dl (Grosse et al. 2002).

[26] Recall that the children in the testing sample are 72 months and younger, so a three-year-old is near the middle of this age range.

[27] In 2015, approximately 1 million children aged 4 and younger lived in race counties, 3 million lived in border counties, and 15 million lived in control counties.

[28] If our estimates are mapped to the full population, the effect on elevated blood lead rates in race counties would result in approximately an extra 100 poisoned children in the average race county.

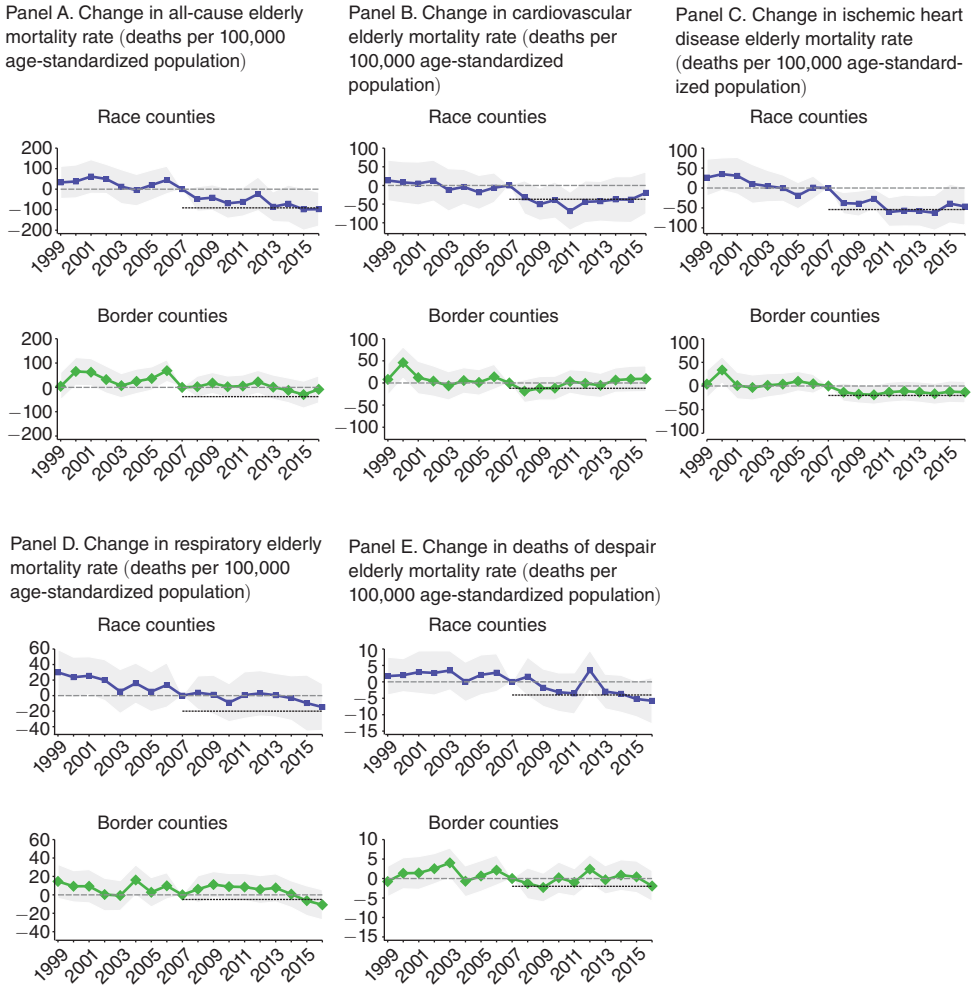



FIGURE 7. THE EFFECT OF LEADED RACES ON ELDERLY MORTALITY RATES IN RACE AND BORDER COUNTIES

*Notes:* Panel A shows estimates for all-cause mortality; panel B shows estimates for cardiovascular mortality; panel C shows estimates for ischemic heart disease (IHD) mortality; panel D shows estimates for respiratory mortality; and panel E shows estimates for deaths of despair (Case and Deaton 2015). The top panel of each subfigure reports coefficients (blue squares) for race counties, and the bottom panel reports coefficients (green diamonds) for border counties. Each coefficient represents the effect on the age-standardized, elderly morality rate for that cause of death from being in a particular county type relative to 2007, which is omitted. The regression includes state-by-year fixed effects, county fixed effects, and controls for the unemployment rate, median income, percent non-white, tons of lead emitted from TRI facilities, and total manufacturing payroll. All coefficients in each subfigure come from the same regression. The elderly population of the entire United States in the year 2000 was used as the reference population for age standardization. Each regression is weighted by the square root of the elderly population. The shaded gray areas denote the 95 percent confidence interval calculated from robust standard errors clustered at the county level. The black dashed line is the average effect of deleading from our preferred difference-in-difference regression, where the race county treated group consists of those counties that had at least one leaded race prior to 2007, and the border county treated group consists of those counties that did not have a leaded race but bordered a county with a leaded race prior to 2007. This regression defines the post-period as 2007 and after.

This valuation has several limitations. First, we do not have data on actual blood lead levels so we do not directly observe how they change following deleading. Second, this valuation does not account for how long each child's blood lead was elevated and how this length may matter for IQ or future earnings. Finally, because lead is only detectable in the blood stream for a month following exposure, the procedure implicitly assumes that once blood lead returns to baseline levels, the IQ effects and associated future earnings losses also dissipate. This is in contrast to physiological evidence that lead exposure can permanently damage neurological pathways and development.

### B. *Elderly Mortality*

Since the effects of leaded races on elderly mortality are acute and permanent, the calculation of their dollar value is more straightforward and we use them to lower bound the social cost of leaded gasoline. We first use the difference-in-difference estimates shown in Figure 7, which indicate that race counties have higher all-cause elderly mortality rates by 91 deaths per 100,000 during leaded race years, and border counties have all-cause mortality rates higher by 38 deaths per 100,000. We then compute the number of premature deaths by combining the mortality estimates with data on the population over 65.[29] In total, our estimates suggest that leaded races caused 32,000 premature elderly deaths between 1999 and 2006. Next, we translate the count of premature deaths into dollar terms. The average age of a person over 65 is 74, and the expected age of death is 87 conditional on reaching 74 (Murphy et al. 2017). Following Deryugina et al. (2019), we deflate the number of life-years lost by 31 percent to account for possible harvesting (i.e., that those who are likely to die prematurely from pollution are also more likely to die prematurely from other causes). We value each life year lost at approximately $140,000 in 2019 dollars (Cutler 2004). On average, each premature elderly death from lead exposure generates approximately $550,000 in social costs, and total annual elderly mortality externalities are approximately $2.2 billion per year.

To compute the social cost of lead in dollars per gram, we normalize the annual mortality cost by an estimate of the annual lead emissions. We estimate annual lead emissions to be approximately 2 million grams in online Appendix Section A.11. Combining these two estimates, the social cost of a gram of lead added to racing fuel is approximately $1,100.

## V. Conclusion

The main objective of this research is to causally estimate the effect of leaded gasoline on elderly mortality. We do so by exploiting both regulatory exemptions and a natural experiment that together serve to separately identify the effect of leaded gasoline from other potential confounders. The importance of our results extends beyond ascribing a value to automotive racing fuel decisions. Our findings indicate

[29] In 2015, approximately 3 million elderly lived in race counties, 7 million lived in border counties, and 36 million lived in control counties.

that even at present-day levels of overall lead exposure, additional exposure can generate heightened mortality risk. The estimated relationship between lead exposure and these outcomes is both statistically significant and economically meaningful. Our estimates likely understate the true magnitude of the mortality effect of leaded gasoline. Due to the structure of the data and analysis, our mortality analysis is at the county-year level, which introduces measurement error that may attenuate our estimates.

The mortality estimates place a lower bound on the social cost of each gram of tetraethyl lead added to gasoline at roughly $1,100. This estimate is conditional on several factors. First, since lead emissions from races affect outcomes through changes in ambient concentrations, the estimated social cost is conditional on the ambient levels of lead during the time-frame of our mortality analysis: 1999 to 2016. Ambient levels during this time were much lower than in earlier decades, and they continue to slowly decline. This matters for computing the social cost of lead if, as prior work suggests, the dose response is nonlinear with respect to ambient concentrations (e.g., Grönqvist, Nilsson, and Robling 2017). Second, the social cost estimate is the average marginal effect over the range of quantities of lead emitted across all races. The marginal effect of the first gram emitted by a race may be much different than the marginal effect of the 100,000th gram emitted near the end of a multirace weekend. If the lead dose-response function is nonlinear and ambient concentrations continue to decline, then understanding the marginal effect near the origin of the dose-response function is critical for future policy design.

An important issue not studied here is the lack of information about lead in gasoline. To the best of our knowledge, there is no public information on the quantity or location of leaded racing fuel sales. Depending on the location and timing of its use, the social cost of a particular gallon of leaded gasoline may vary dramatically. There is also little information available on the lead content of unleaded gasoline. Although its name implies the absence of lead, unleaded gasoline simply contains no added lead. Lead, in addition to other toxic heavy metals like cadmium, is naturally found in crude oil and may persist in trace amounts after the refining process. Lead can enter the unleaded fuel supply from storage tanks, pipes, solder, or through cross-contamination from use of the same pipelines for leaded aviation gasoline. More information on leaded gasoline sales and the lead content of unleaded fuel is critical for determining social costs and the appropriate amount of regulatory scrutiny.

Our findings shed light on the consequences of the continued use of leaded gasoline for racing and aviation. Our mortality cost estimate indicates that the social cost of leaded racing fuel is substantially higher than the private costs reflected in current market prices. While we do not directly study leaded aviation fuel, our estimates are informative of the external costs associated with its use. Even if the external effects of aviation gasoline are only 1 percent ($10 per gram) of our racing fuel estimates, then the socially efficient price of leaded aviation fuel would be three times higher than current market prices. This suggests that there are significant global damages from leaded aviation gasoline.

Our findings also have implications for current lead poisoning policies. Much of the concern about lead is heavily focused on children as evidenced by the CDC's nationwide Childhood Lead Poisoning Prevention Program as well as numerous state- and county-level efforts. Our results demonstrate the large costs imposed on the elderly and the potential benefits of expanding lead exposure mitigation efforts to include this overlooked sensitive group.

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